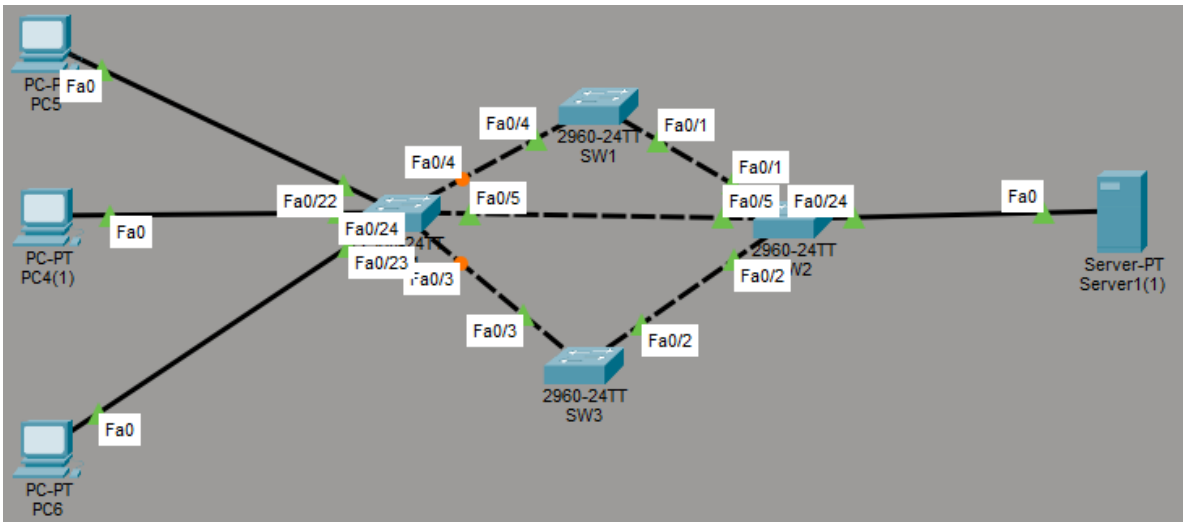
REPORTE DE ACTIVIDAD #2 – ETHERCHANNEL DE CAPA 2

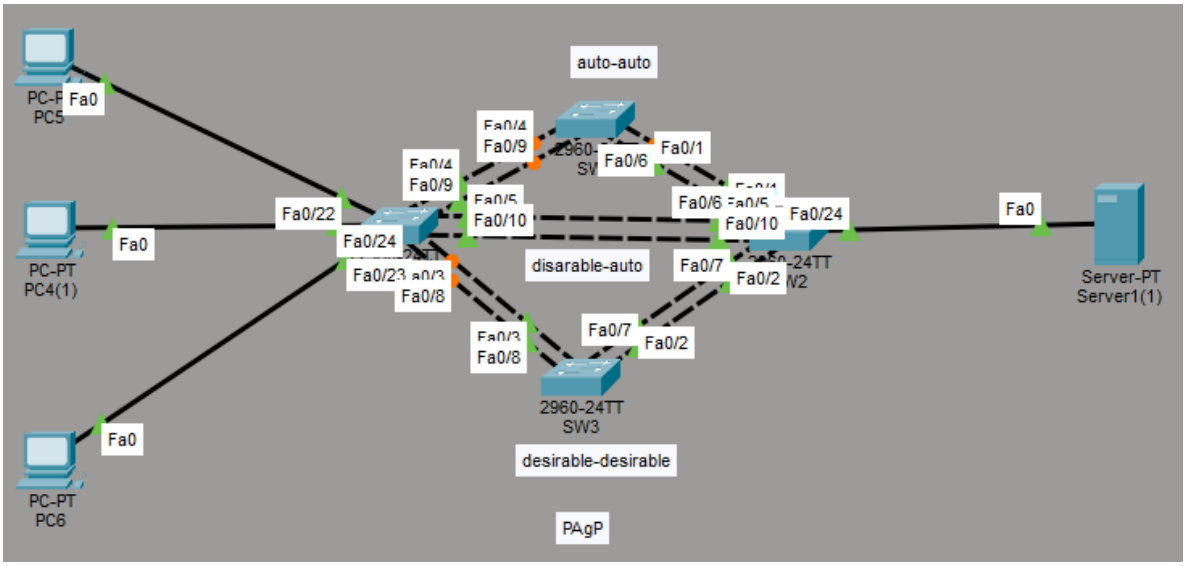
Port Aggregation Protocol

Topología inicial



Comenzamos con una topología simple: un conjunto de computadoras que se conectan a un servidor a través de una red de varios switches. A pesar de ser estable y tener algo de redundancia, el ancho de banda es limitado y quedan muchos puertos disponibles sin usar en los switches.

Topología final

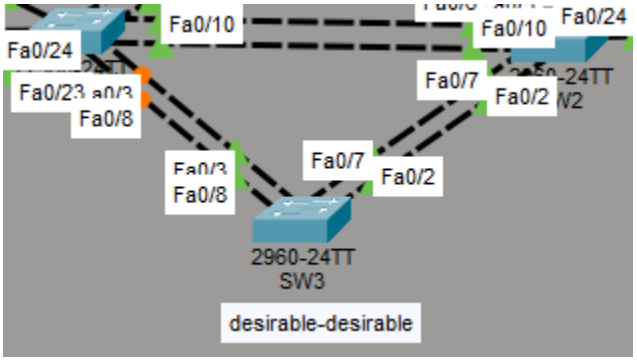


Ahora hay dos enlaces físicos por cada conexión entre cada switch, aumentando el ancho de banda sin necesidad de invertir en interfaces físicas costosas. Veamos el cómo.

Guía paso a paso

Para demostrar los métodos de negociación de canales en PAgP, haremos tres conexiones distintas por cada combinación de modos de negociación.

desirable-desirable



En este método ambos extremos del enlace lógico intentan iniciar la negociación.

En SW4:

```
interface Fa0/3
shutdown
channel-group 1 mode desirable
no shutdown
interface Fa0/8
shutdown
channel-group 1 mode desirable
no shutdown
```

Ingresamos a la configuración de cada interfaz que se agregará al canal. En este caso las interfaces 3 y 8 FastEthernet se agregarán al canal 1 en modo desirable.

En SW3:

```
interface Fa0/3
shutdown
channel-group 1 mode desirable
no shutdown
interface Fa0/8
shutdown
channel-group 1 mode desirable
no shutdown

interface Fa0/2
shutdown
channel-group 4 mode desirable
no shutdown
interface Fa0/7
shutdown
channel-group 4 mode desirable
no shutdown
```

Hacemos exactamente lo mismo, agregar los puertos 3 y 8 al canal 1 en modo desirable para que ambos lados inicien la negociación, además de agregar los puertos 2 y 7 al canal 4 también en modo desirable.

En SW2:

```
interface Fa0/2
shutdown
channel-group 4 mode desirable
no shutdown
interface Fa0/7
shutdown
channel-group 4 mode desirable
no shutdown
```

Por último agregamos las interfaces 2 y 7 al canal 4 para iniciar la negociación con el SW3.

Ahora veamos el estado de los canales creados:

```
SW4#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           PAgP    Fa0/3(P) Fa0/8(P)
```

```
SW3#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           PAgP    Fa0/3(P) Fa0/8(P)
4      Po4(SU)           PAgP    Fa0/2(P) Fa0/7(P)
```

```
SW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

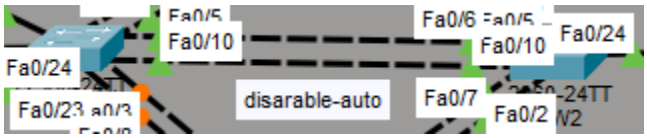

Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------


4      Po4(SU)           PAgP    Fa0/2(P) Fa0/7(P)
```

Observamos que en los tres switches los dos canales creados se encuentran en estado “SU”, que significa que fueron creados exitosamente en capa 2.

desirable-auto



En este modo un extremo inicia la negociación y el otro lo espera. En este paso, SW2 estará esperando la negociación y SW4 la iniciará.

En SW4:

```
interface Fa0/5
shutdown
channel-group 2 mode desirable
no shutdown
interface Fa0/10
shutdown
channel-group 2 mode desirable
no shutdown
```

Se agregan los puertos 5 y 10 al grupo 2 en modo desirable para iniciar la negociación.

En SW2:

```
interface Fa0/5
shutdown
channel-group 2 mode auto
no shutdown
interface Fa0/10
shutdown
channel-group 2 mode auto
no shutdown
```

Se incluyen los puertos 5 y 10 al canal 2 en modo auto para esperar la negociación.

Veamos el resultado:

```
SW4#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           PAgP   Fa0/3(P) Fa0/8(P)
2      Po2(SU)           PAgP   Fa0/5(P) Fa0/10(P)
```

```
SW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

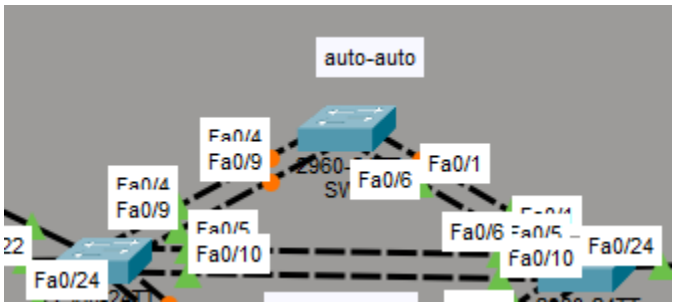

Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

2      Po2(SU)           PAgP   Fa0/5(P) Fa0/10(P)
```

El canal 2 se encuentra en estado “SU”, por lo que se ha montado exitosamente.

auto-auto



Aquí ambos extremos estarán esperando la negociación, por lo que el canal nunca se formará.

En SW4:

```
interface Fa0/4
shutdown
channel-group 3 mode auto
no shutdown
interface Fa0/9
shutdown
channel-group 3 mode auto
no shutdown
```

Se incluyen las interfaces 4 y 9 al grupo 3 en modo auto para esperar la negociación.

En SW1:

```
interface Fa0/1
shutdown
channel-group 5 mode auto
no shutdown
interface Fa0/6
shutdown
channel-group 5 mode auto
no shutdown

interface Fa0/4
shutdown
channel-group 3 mode auto
no shutdown
interface Fa0/9
shutdown
channel-group 3 mode auto
no shutdown
```

Se agregan las interfaces 1 y 6 al canal 5 y las 4 y 9 al canal 3 esperando la negociación.

En SW2:

```
interface Fa0/1
shutdown
channel-group 5 mode auto
no shutdown
interface Fa0/6
shutdown
channel-group 5 mode auto
no shutdown
```

Los puertos 1 y 6 en el canal 5 esperan la negociación.

De este modo ningún puerto inicia la negociación, por lo que los canales no se formarán en ningún switch, como se puede ver:

```
SW4#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           PAgP   Fa0/3(P) Fa0/8(P)
2      Po2(SU)           PAgP   Fa0/5(P) Fa0/10(P)
3      Po3(SD)           PAgP   Fa0/4(I) Fa0/9(I)
```

```
SW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

3      Po3(SD)           PAgP   Fa0/4(I) Fa0/9(I)
5      Po5(SD)           PAgP   Fa0/1(I) Fa0/6(I)
```

```
SW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

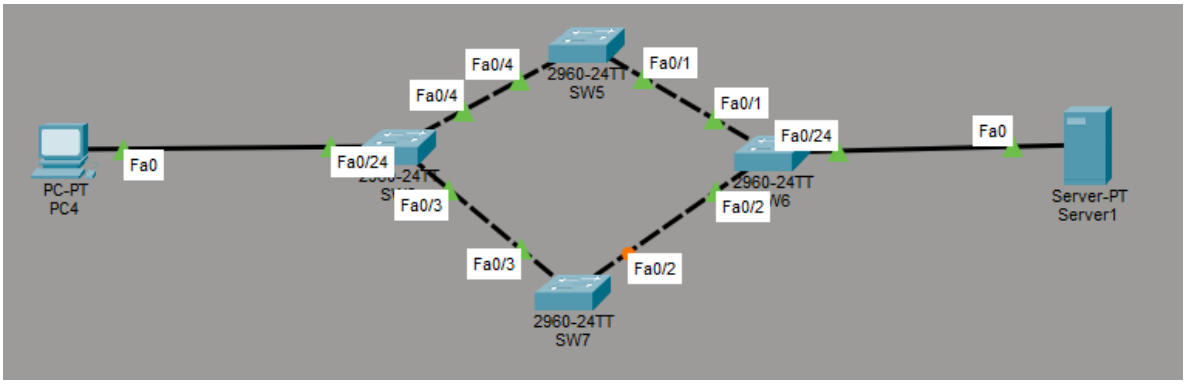
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

2      Po2(SU)           PAgP   Fa0/5(P) Fa0/10(P)
4      Po4(SU)           PAgP   Fa0/2(P) Fa0/7(P)
5      Po5(SD)           PAgP   Fa0/1(I) Fa0/6(I)
```

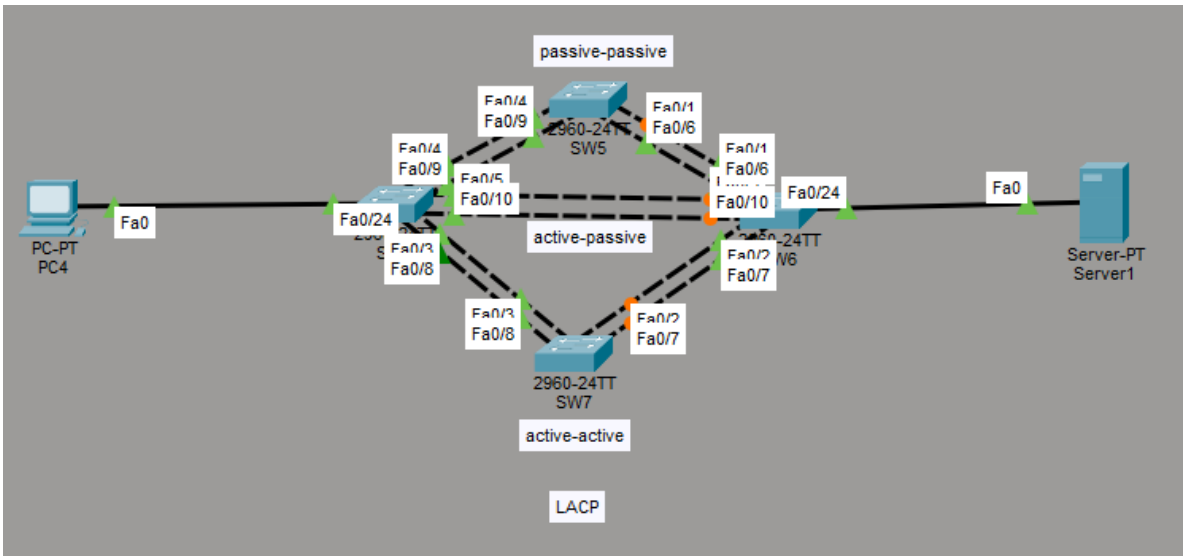
Los canales 1 y 5 en los tres switches se encuentran en modo “SD”, que significa que se incluyeron en un canal de capa 2 pero este canal nunca se formó.

Link Aggregation Control Protocol

Topología inicial



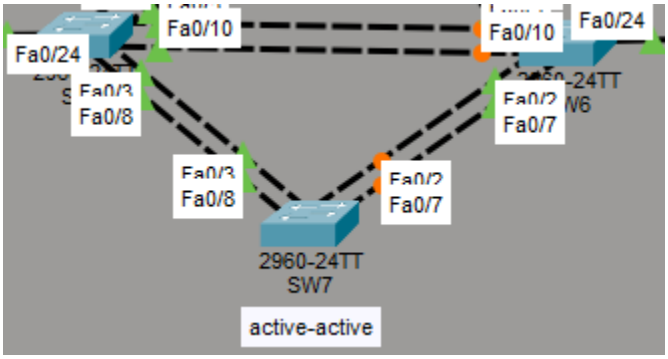
Topología final



En vez de usar un protocolo propietario de Cisco, utilizaremos LACP, un protocolo de estándar abierto. Además agregaremos un enlace extra para demostrar los tres modos de LACP.

Guía paso a paso

active-active



Ambos extremos iniciarán activamente la negociación.

En SW8:

```
interface Fa0/3
shutdown
channel-group 1 mode active
no shutdown
interface Fa0/8
shutdown
channel-group 1 mode active
no shutdown
```

Agregamos los puertos 3 y 8 en modo active al canal 1.

En SW7:

```
interface Fa0/3
shutdown
channel-group 1 mode active
no shutdown
interface Fa0/8
shutdown
channel-group 1 mode active
no shutdown

interface Fa0/2
shutdown
channel-group 4 mode active
no shutdown
interface Fa0/7
shutdown
channel-group 4 mode active
no shutdown
```

Se incluyen los puertos 3 y 8 al canal 1 y el 2 y 7 al canal 4 en modo active.

En SW6:

```
interface Fa0/2
shutdown
channel-group 4 mode active
no shutdown
interface Fa0/7
shutdown
channel-group 4 mode active
no shutdown
```

Se agregan los puertos 2 y 7 en modo active al canal 4.

Veamos los resultados:

```
SW8#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           LACP   Fa0/3(P) Fa0/8(P)
```

```
SW7#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           LACP   Fa0/3(P) Fa0/8(P)
4      Po4(SU)           LACP   Fa0/2(P) Fa0/7(P)
```

```
SW6#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

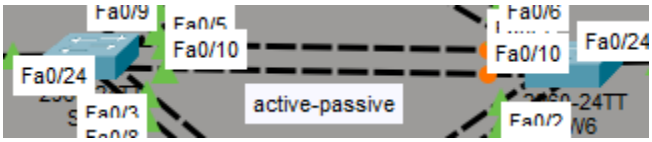

Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------


4      Po4(SU)           LACP   Fa0/2(P) Fa0/7(P)
```

Los dos canales en los tres switches se encuentran en estado “SU”, creados exitosamente.

active-passive



El extremo de SW8 iniciará activamente la negociación y SW6 la esperará.

En SW8:

```
interface Fa0/5
shutdown
channel-group 2 mode active
no shutdown
interface Fa0/10
shutdown
channel-group 2 mode active
no shutdown
```

Los puertos 5 y 10 se agregan en modo active al canal 2.

En SW6:

```
interface Fa0/5
shutdown
channel-group 2 mode passive
no shutdown
interface Fa0/10
shutdown
channel-group 2 mode passive
no shutdown
```

Se incluyen los puertos 5 y 10 en modo passive al grupo 2.

Vemos que el canal se forma de manera exitosa:

```
SW8#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------

1      Po1(SU)           LACP   Fa0/3(P) Fa0/8(P)
2      Po2(SU)           LACP   Fa0/5(P) Fa0/10(P)
```

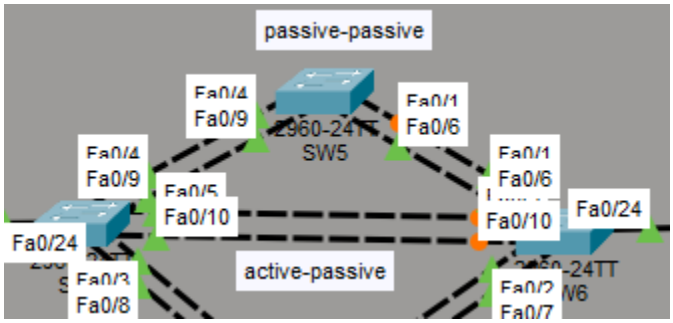
```
SW6#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------

2      Po2(SU)           -      Fa0/5(P) Fa0/10(P)
4      Po4(SU)           LACP   Fa0/2(P) Fa0/7(P)
```

passive-passive



Ambos lados esperan la negociación, evitando la formación del canal.

En SW8:

```
interface Fa0/4
shutdown
channel-group 3 mode passive
no shutdown
interface Fa0/9
shutdown
channel-group 3 mode passive
no shutdown
```

Se agregan las interfaces 4 y 9 al canal 3 en modo passive.

En SW5:

```
interface Fa0/1
shutdown
channel-group 5 mode passive
no shutdown
interface Fa0/6
shutdown
channel-group 5 mode passive
no shutdown

interface Fa0/4
shutdown
channel-group 3 mode passive
no shutdown
interface Fa0/9
shutdown
channel-group 3 mode passive
no shutdown
```

Se incluyen las interfaces 1 y 6 al canal 5 y la 4 y la 9 al canal 3 en modo passive.

En SW6:

```
interface Fa0/1
shutdown
channel-group 5 mode passive
no shutdown
interface Fa0/6
shutdown
channel-group 5 mode passive
no shutdown
```

Agregamos los puertos 1 y 6 al canal 5 en modo passive.

Vemos que el canal no se abrió:

```
SW8#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           LACP   Fa0/3(P) Fa0/8(P)
2      Po2(SU)           LACP   Fa0/5(P) Fa0/10(P)
3      Po3(SD)           LACP   Fa0/4(D) Fa0/9(D)
```

```
SW5#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

3      Po3(SD)           LACP   Fa0/4(D) Fa0/9(D)
5      Po5(SD)           LACP   Fa0/1(D) Fa0/6(D)
```

```
SW6#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

2      Po2(SU)           -      Fa0/5(P) Fa0/10(P)
4      Po4(SU)           LACP   Fa0/2(P) Fa0/7(P)
5      Po5(SD)           LACP   Fa0/1(D) Fa0/6(D)
```